Лекция4

24 февраля 2026 г. 12:18

$$-\left(\frac{\partial p}{\partial S}\right)_V = \left(\frac{\partial T}{\partial V}\right)_{S}$$

$$\left(\frac{\partial V}{\partial S}\right)_p = \left(\frac{\partial T}{\partial p}\right)_{S}$$

$$dU = TdS + fdl$$

$$F = U - TS; \quad dF = -SdT + fdl$$

$$\left(\frac{\partial l}{\partial T}\right)_f = -\left(\frac{\partial l}{\partial f}\right)_T \left(\frac{\partial f}{\partial T}\right)_l = \underset{>0}{\left(\frac{\partial l}{\partial f}\right)_T} \left(\frac{\partial S}{\partial l}\right)_T \ominus$$

$$-\left(\frac{\partial S}{\partial l}\right)_T = \left(\frac{\partial f}{\partial T}\right)_l$$

$$\left(\frac{\partial S}{\partial l}\right)_T = -\left(\frac{\partial S}{\partial T}\right)_l \left(\frac{\partial T}{\partial l}\right)_S$$

$$\frac{C_l}{T} > 0$$

$$\ominus \; -\frac{C_l}{T}\left(\frac{\partial l}{\partial f}\right)_T \left(\frac{\partial T}{\partial l}\right)_S < 0$$

$$\left(\frac{\partial T}{\partial l}\right)_S > 0$$

Как не запоминать соотношение Максвелла (Якобиан)

$$\frac{\partial(f,g)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial g} \end{vmatrix} = \left(\frac{\partial f}{\partial x}\right)_y \left(\frac{\partial g}{\partial y}\right)_x - \left(\frac{\partial f}{\partial y}\right)_x \left(\frac{\partial g}{\partial x}\right)_y$$

1) $\frac{\partial(f,y)}{\partial(x,y)} = \left(\frac{\partial f}{\partial x}\right)_y$

2) Если поменять параметры местами, то изменится знак. $\frac{\partial(f,g)}{\partial(x,y)} = -\frac{\partial(g,f)}{\partial(x,y)}$

3) $\frac{\partial(f,g)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(\alpha,\beta)} = \frac{\partial(f,g)}{\partial(\alpha,\beta)}$

$$\left(\frac{\partial P}{\partial S}\right)_V = \frac{\partial(P,V)}{\partial(S,V)}$$

$$\left(\frac{\partial T}{\partial V}\right)_S = \frac{\partial(T,S)}{\partial(V,S)} = -\frac{\partial(T,S)}{\partial(S,V)} \Rightarrow \frac{\partial(T,S)}{\partial(P,V)} = 1$$

$$\partial(T,S) = \partial(P,V)$$

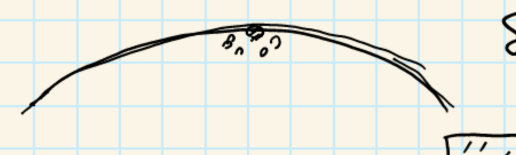
$$\left(\frac{\partial V}{\partial S}\right)_P = \frac{\partial(V,P)}{\partial(S,P)} = \frac{\partial(S,T)}{\partial(S,P)} = \left(\frac{\partial T}{\partial P}\right)_S$$

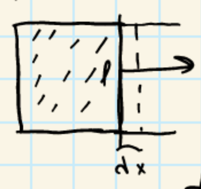
$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{\partial(P,V)}{\partial(T,V)} = \frac{\partial(T,S)}{\partial(T,V)} = \left(\frac{\partial S}{\partial V}\right)_T$$

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{\partial(V,P)}{\partial(T,P)} = \frac{\partial(S,T)}{\partial(T,P)} = -\left(\frac{\partial S}{\partial P}\right)_T$$

Поверхностное натяжение

$$\delta A = \sigma d\Pi$$





$\boxed{f = \sigma l}$

$F = F_{об} + F_{пов}$

$F_{пов} = \sigma \cdot \Pi$

$\sigma > 0$ – пов. натяжение

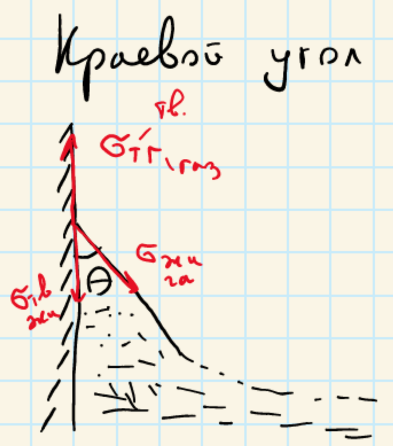
$f dx = dF = \sigma d\Pi = \sigma l dx$

$$dF_{пов} = -S_{пов}dT + \underset{\delta A_T}{\sigma d\Pi}; \quad F_{пов} = \sigma(T) \cdot \Pi$$

$$S_{пов} = -\left(\frac{\partial F_{пов}}{\partial T}\right)_\Pi = -\frac{d\sigma}{dT} \cdot \Pi$$

$$\delta Q = TdS_{пов} = \underbrace{-T\frac{\partial S}{\partial T}}_{\text{теплота образования пов-ти}} \cdot \Delta\Pi$$

Краевой угол



$\sigma_{тг} = \sigma_{тж} + \sigma_{жг}\cos\Theta$

$$\cos\Theta = \frac{\sigma_{тг} - \sigma_{тж}}{\sigma_{жг}}$$

$\Theta = 0$ – полное смачивание

$\Theta = \pi$ – полное несмачивание – не бывает

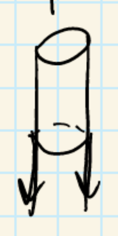
Давление Лапласа

$P_1 \cdot 2r \cdot l = P_2 \cdot 2rl - 2\sigma l$

$$P_2 - P_1 = \frac{\sigma}{r}$$

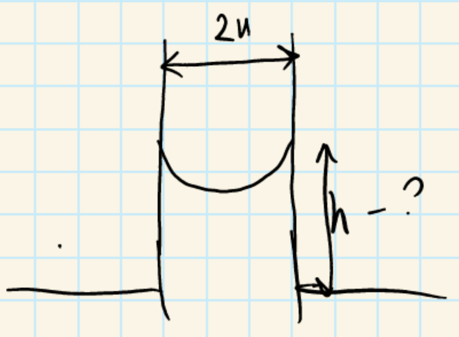
$P_1 =$

$$\frac{P_1 - P_2}{P_1} = \frac{\sigma}{\nu r_1}$$



$$P_2 - P_1 = \frac{2\sigma}{r}$$

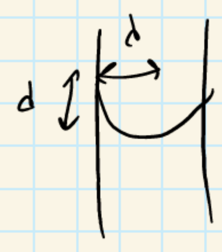
Подъём в капилляре



$$\rho g h = \frac{2\sigma}{r}$$

$$h = \frac{2\sigma}{\rho g r}$$

Капиллярные постоянные



$$d = \frac{\sigma}{\rho g d/2} \Rightarrow d^2 \sim \frac{\sigma}{\rho g} \quad \text{для воды} \sim 3 \text{ мм}$$